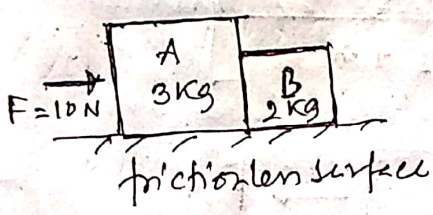
# Problem on Newton's Laws of Motion

1. A car of mass 100 kg is moving with a velocity 2m/s. What force must be applied to the car to stop it within a distance of 10m. [20N]

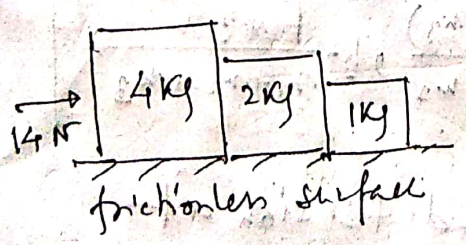
2. A cricket player catches a ball of mass 0.1kg moving with a speed of 10m/s in 0.1 sec. Find force exerted by him. [10N]

3. A balloon of mass 'm' is going downward with constant acc$^n$ 'a'. What mass of sand bags should be dropped from the balloon so that it rises with same acceleration. [Neglect air resistance] $\left[\frac{2ma}{g+a}\right]$

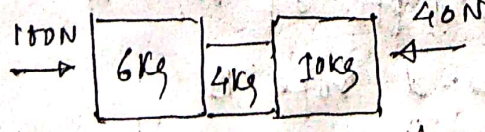
4. 


(i) In the fig how much force does block A apply on block B? (4N)

(ii) If the 10N force (horizontal) were applied on block B towards block A, how much force would block B have applied on block A? (6N)

5. 


Find contact force between blocks 4kg and 2kg (6N)

6. 


(i) Find acc$^n$ of the system and force of contact betn (ii) 6kg and 4kg. (iii) & that betn 4kg and 10kg blocks [(i) 3 m/s$^2$ (ii) 82 N (iii) 70N]

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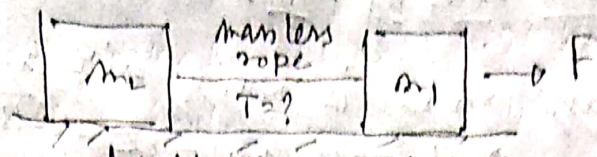
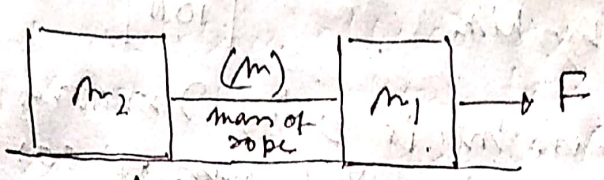
7.




Two blocks $m_1$ and $m_2$ are connected by a very light rope. Force F is applied on $m_1$. Find tension in rope $\left[\frac{m_2 F}{m_1+m_2}\right]$

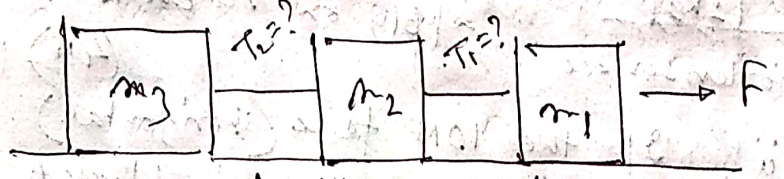
8.




mass of rope connected betn $m_1$ and $m_2$ is 'm'. Force F is applied on $m_1$. Find tension at the mid pt of rope

$$\left[\frac{F\left(m_2+\frac{m}{2}\right)}{m_1+m_2+m}\right]$$

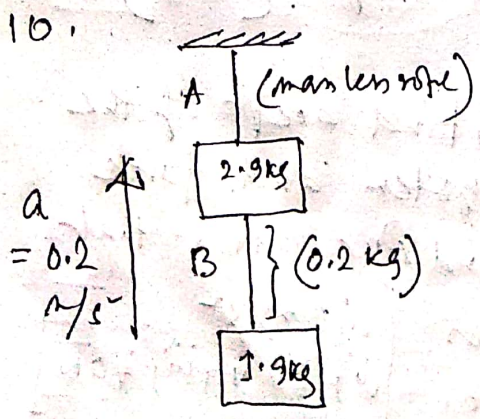
9.




Find tension in the massless ropes betn. (i) $m_1$ and $m_2$ (ii) & betn. $m_2$ and $m_3$

$$\left[T_1=\frac{F(m_2+m_3)}{m_1+m_2+m_3},\ T_2=\frac{F m_3}{m_1+m_2+m_3}\right]$$

10.




Find tension at mid pts of rope A and rope B

[50N, 20N]

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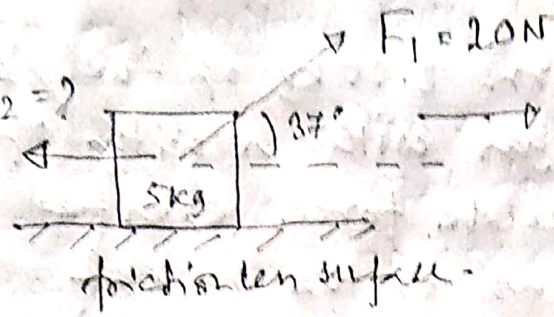



Find $F_2 = ?$

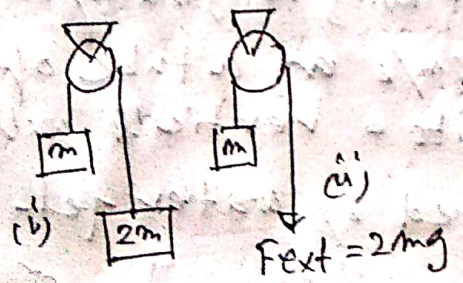
[Ans 2N]

12. A cricket ball of mass 0.25 kg moving with speed 10 m/s collides with a bat and returns with same speed 10 m/s within 0.01 sec. Find force exerted by bat on ball? (500 N)

13. A mass 1 kg is suspended by a thread. It is (i) lifted with an acc$^n$ of 5 m/s² (ii) lowered with acc$^n$ of 5 m/s². Find ratio of tensions in thread in two cases. (3:1)
($g = 10$ m/s²)

14. A rope can withstand (সহ্য করতে পারে) a tension of 10,000 N. What is the max$^m$ acc$^n$ with which the rope can raise a weight 800 kg? (2.45 m/s²)

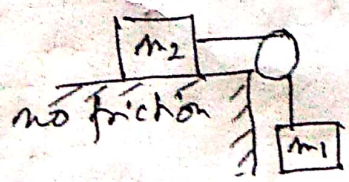
15.




Find acc$^n$ of block of mass m in case (i) & (ii)

[(i) $a = \frac{g}{3}$ ↓
(ii) $a = g$ ↓]

16.




$m_1 > m_2$. Find acc$^n$ of system and tension in rope

[$a = \frac{m_1 g}{m_1 + m_2}$

$T = \frac{m_1 m_2 g}{m_1 + m_2}$]

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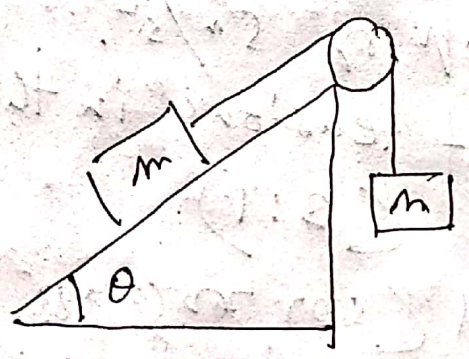
17. A rocket burns 0.25 kg of fuel/sec and ejects a gas with velocity of 1 km/sec. Calculate force exerted by ejected gas on the rocket (250 N)

~~and~~ ~~up~~

18. A body of mass 10 kg falls from a height of 20 m on horizontal surface. If resistance offered by the surface is $10^4$ N, then how far will the body penetrate into that surface ($g = 10\ m/s^2$) [Ans = 0.4 m or 40 cm]

19.




Find accⁿ of the system of blocks (in fig) connected by massless string passing over a smooth pulley. (all surfaces are frictionless)

Ans $\left[a = \frac{g}{2}(1 - \sin\theta)\right]$

20. A jet of water flowing horizontally with a speed of 15 m/s as a tube of cross section $10^{-2}\ m^2$ hits a vertical wall. What force is exerted by water jet on wall. [2250 N]

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